



Optimization of digital image processing to determine quantum dots' height and density from atomic force microscopy



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ARTICLE INFO

Article history:

Received 19 January 2017

Revised 26 April 2017

Accepted 22 September 2017

Available online 23 September 2017

Keywords:

Atomic force microscopy

Quantum dots

Image processing

Local threshold

ABSTRACT

An optimized method of digital image processing to interpret quantum dots' height measurements obtained by atomic force microscopy is presented. The method was developed by combining well-known digital image processing techniques and particle recognition algorithms. The properties of quantum dot structures strongly depend on dots' height, among other features. Determination of their height is sensitive to small variations in their digital image processing parameters, which can generate misleading results. Comparing the results obtained with two image processing techniques – a conventional method and the new method proposed herein – with the data obtained by determining the height of quantum dots one by one within a fixed area, showed that the optimized method leads to more accurate results. Moreover, the log-normal distribution, which is often used to represent natural processes, shows a better fit to the quantum dots' height histogram obtained with the proposed method. Finally, the quantum dots' height obtained were used to calculate the predicted photoluminescence peak energies which were compared with the experimental data. Again, a better match was observed when using the proposed method to evaluate the quantum dots' height.

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1. Introduction

Quantum dots (QDs) can, in principle, improve the performance of different optoelectronic devices such as lasers [1], photodetectors [2] and solar cells [3]. It is well known that the properties of these nanometric structures strongly depend on their shape, size and composition [4], therefore a morphological characterization of the QDs is imperative. In the design, manufacture and optimization of devices based on QDs, it is then fundamental to properly and as accurately as possible characterize them. However, this may present a problem since the measurements are on a nanometer scale, where a small error in the dimensions can lead to major differences in device performance. Therefore, it is important to develop a reliable method to morphologically characterize the QDs. Atomic Force Microscopy (AFM) 3D images have been extensively used to determine the height and areal density of surface QDs [5,6]. The AFM images are usually digitally processed using a commercial software, which usually comes with the equipment, to

determine the sample characteristics but, depending on the image analysis, different results can be obtained from the same AFM image.

In this work, we propose an optimized method to digitally process the AFM images to more accurately determine both the height and density of the QDs.

2. Experimental details

The investigated samples have the following structure: an InP buffer layer is deposited on a (100) oriented InP substrate followed by a layer of quaternary material (InGaAlAs) lattice-matched to the substrate. Both layers are grown at 630 °C. After lowering the temperature to 520 °C, the InAs QDs are nucleated on this surface. After 12 s at this temperature the 10 nm InP capping layer is deposited while the temperature is ramped up. Then, a second layer of the quaternary material is grown at 630 °C, on which the free standing QDs are deposited. A scheme of the sample is shown in Fig. 1. The QDs are nucleated according to the Stranski-Krastanov growth mechanism where, prior to the development of the quantum dots, a wetting layer is formed [7].

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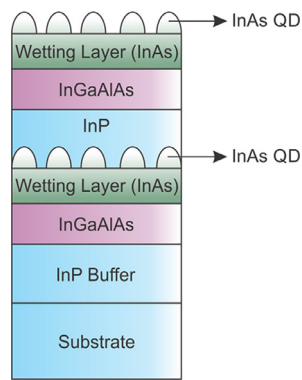


Fig. 1. Sample structure.

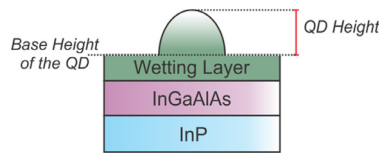


Fig. 2. Detail of QD height.

The samples were grown by metalorganic vapor phase epitaxy (MOVPE), a method which can be used under various conditions, leading to different QD size distributions. The optical and morphological properties of QD samples can be characterized with different techniques.

The AFM measurements were performed on a NX-10 scanning probe microscope (Park Systems). The microscope was operated in non-contact mode (NC-AFM), using a steep silicon tip, with a nominal radius of 8 nm. In this method, the tip, which is mounted on a cantilever, oscillates near its resonant frequency. As the tip scans the samples' surface, attractive forces between the tip and the sample causes a shift in the amplitude and frequency of oscillation of the cantilever. The AFM feedback system was set to track the changes in the amplitude, adjusting the tip-sample distance to maintain it at a constant value.

In our experiment, the tip was set to oscillate at a driving frequency of ~ 76 kHz. Images were taken at several different locations, at a scan rate of 1 Hz, with scanning sizes of $2\mu\text{m} \times 2\mu\text{m}$ and $1\mu\text{m} \times 1\mu\text{m}$ with 1024 lines. Resulting topographic images were plane processed, and all images were taken at ambient air with $\sim 45\%$ RH and at room temperature $\sim 23^\circ\text{C}$.

3. Description of the problem

The proper way to determine the height of a QD from AFM images is to obtain the difference between its maximum and base heights, as shown in Fig. 2.

Most technological applications require a high QD density. Therefore, a one by one height analysis of a large number of QDs is an inefficient and time-consuming process. Thus, image processing methods are available to automate this procedure. The widely used WSxM software [8] contains the so-called *flooding* method [9] that performs these measurements automatically. However, this method does not estimate the QD height with enough accuracy. The *flooding* method allows one to detect the highest (hills, dots) or lowest (holes, valleys) features of an image. With the most common option (*find hills*), all the hills with height above a given threshold are considered as dots. This threshold is chosen by the user and its value is extremely important because the determined height of each QD directly depends on this choice. The estimated height of each QD is just the difference between the value of the

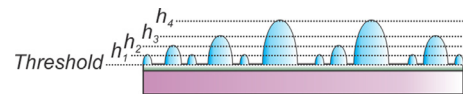


Fig. 3. Scheme of a sample surface with QDs and a process to determine their height.

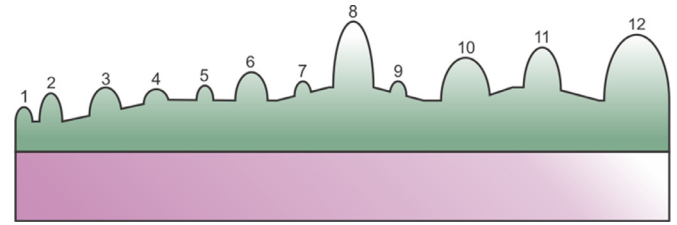


Fig. 4. Scheme of sample surface with QDs on top of an uneven background.

maximum height of each object and the value of its threshold, as schematically represented in Fig. 3.

Actually, the base height of each quantum dot varies across the sample, as depicted in Fig. 4, which is exaggerated for clarity reasons. An AFM image depicting this phenomenon is shown in Fig. 5(a). In this image of a sample surface the intensity of each pixel represents the surface height. In the contrast enhanced image of Fig. 5(a), the variation of the surface height is revealed by the grey scale. The height profiles across two QDs, 1 and 2 in the same figure, are depicted in Fig. 5(b), showing that the QDs have in fact a different base line.

In such situations, selecting a global threshold is not an appropriate strategy. A low threshold, as in Fig. 6(a), leads to an overestimation of many QDs' heights and an underestimation of their density. An intermediate threshold, as in Fig. 6(b), implies in the underestimation of the height of some QDs and an overestimation of others. A high threshold, as Fig. 6(c), underestimates the height of some QDs and excludes many others. Thus, it is impossible to find a single accurate global threshold to accurately evaluate the QDs' heights.

This effect is clearly shown for an AFM image in Fig. 7. The images on the left column show the effect of increasing thresholds in the detection of QDs. The yellow outlines show the detected regions in each case. The plots on the right column are height profiles along the two white lines drawn on the left column image. The filled areas represent the detected QDs for the different threshold values. Fig. 7(a) shows the dot distribution height profile when a very low threshold is used. In this situation many QDs and other regions of the image are considered as a single large object as revealed by region number 4. In this case, just a few QDs are correctly identified (QDs 1, 2 and 3 in the same plot). As the threshold increases (Fig. 7(b) and (c)) more and more QDs are detected, but on the other hand, their heights are not accurately determined, as evidenced by the line plots on the right column.

As an example, in the plot of Fig. 7(b) the height of the QDs identified by numbers 1 to 3 are underestimated, while those for the QDs identified as 4, 5 and 6 are more accurate. The used threshold in this evaluation incorrectly considers region 7 as a single object. By further raising the height of the threshold some dots may be eliminated. This is what is observed in the plot of Fig. 7(d), where the QD between numbers 3 and 4 is unaccounted for. Additionally, in the same figure, the base height for the other QDs is clearly wrong.

4. Description of the proposed method

An alternative to the limitation of the global threshold is the use of an adaptive, or local, threshold. In this approach, a window

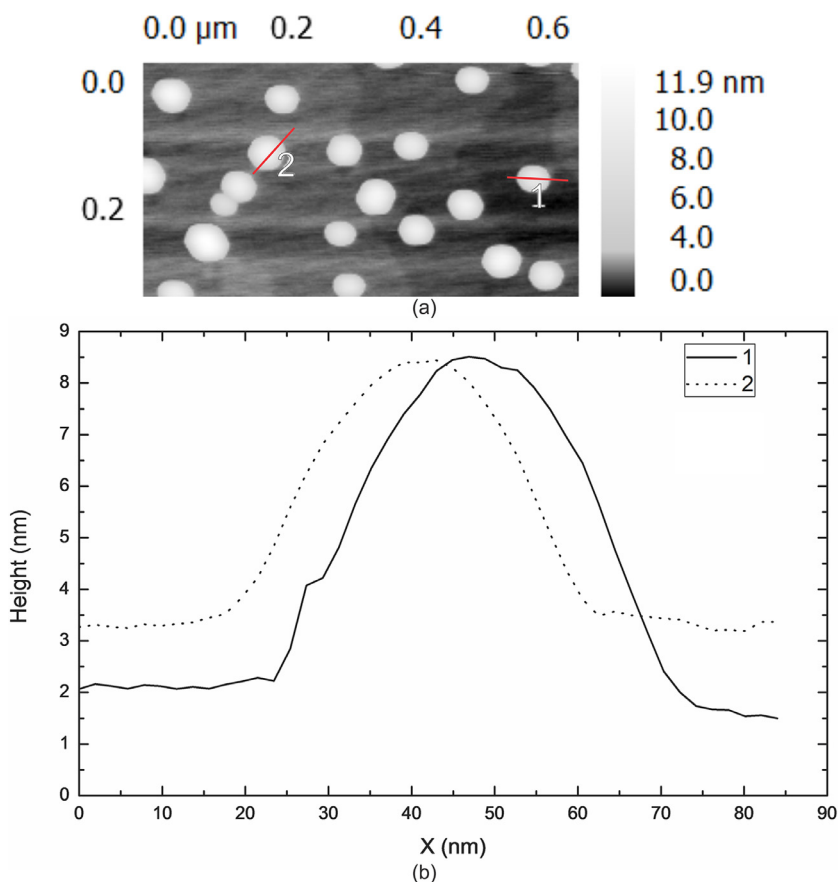


Fig. 5. (a): AFM image of a sample surface with QDs, where the grey scale reveals the surface height. (b): The intensity profile across two selected QDs in the AFM image of Figure 5.a.

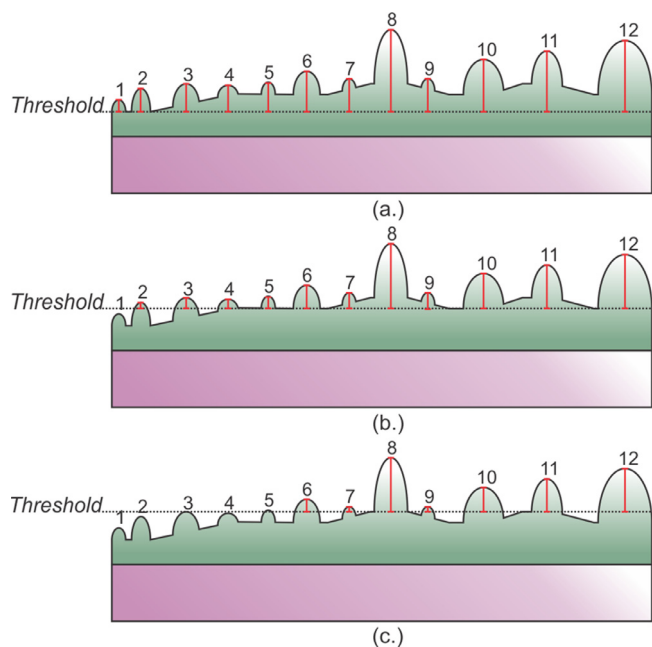


Fig. 6. Scheme, not to scale, of a sample surface with QDs. (a) Low threshold; (b) Intermediate threshold; (c) High threshold.

of adjustable size scans the image, and a threshold is selected for each window position. The local threshold selection can be automated using different kinds of statistical analysis of the pixel intensity distribution in each window. This approach is commonly

used in images that show an uneven background or illumination variation that preclude the use of a global threshold.

In this work we explore the options available under the FIJI [10] software. FIJI is a container for the well-known ImageJ [11] program and automatically configures hundreds of useful plugins. Among those, there is a set of nine Automatic Local Threshold possible routines [12]. These routines scan the image with an analysis window with adjustable radius (R), and dynamically decide if the central pixel of the window is part of an object or the background. The decision criterion is based on various grayscale statistics of pixels within the window. The nine routines are tested many times with different parameters including the window size. Finally, the most appropriate routine is chosen at the end. This selection is based on a comparison between the results of a number of QDs' heights measured manually (references) and the result of the same QDs' heights obtained for each tested routine with many different parameters. The chosen routine is the one that provides the QDs' heights closer to those of the references. The number of reference QDs is determined by each user. The samples investigated presented excellent results with four to five references. Below a short description of these routines follows. Each routine has an equation for segmentation, as shown below. Basically, if the pixel satisfies the condition between parentheses located before the question mark, the pixel is determined as a part of an object, otherwise the pixel is determined as part of the background.

The Median routine [13–16] uses the following equation for segmentation:

$$\text{pixel} = (\text{pixel} > \text{median} - C) ? \text{object} : \text{background}$$

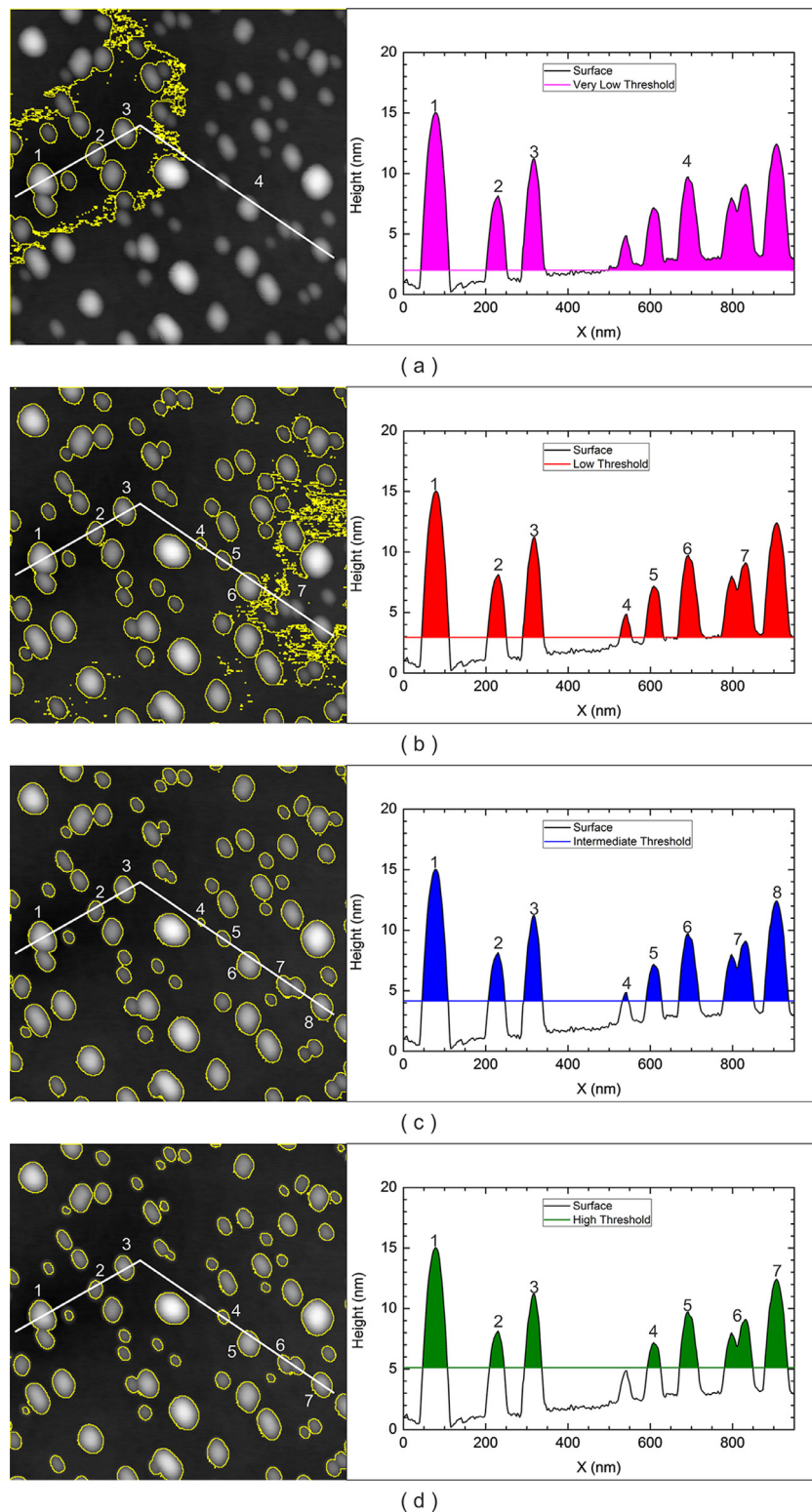


Fig. 7. On the left, an AFM image of a sample surface region with different threshold heights. The yellow edges represent the contours of detected regions. On the right, the height profile along the white lines shown on the left figure. The color filled areas in the plots represent the identified QDs. (a) Very low threshold; (b) Low threshold; (c) Intermediate threshold; (d) High threshold. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

where the median value of the height is calculated for the pixels within each window, as it scans the image, and C is a sensitivity parameter. Thus, a pixel is considered part of the object if it is brighter than $(\text{median} - C)$. As the window moves, the median changes accordingly, and thus a different local threshold is chosen at each location.

The Mean routine [13–16] selects the threshold as the mean of the local greyscale distribution. A variation of this method uses the mean - C , where C is a constant. The parameter 1 is the C value. The default value is 0. Any other number will change the default value. The Mean routine uses the following equation:

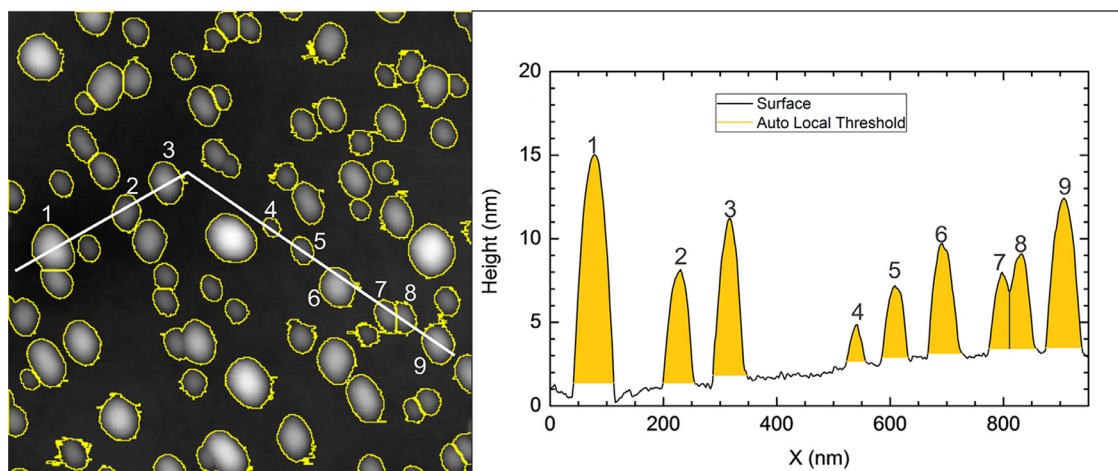


Fig. 8. Median local threshold method applied to an AFM QD image ($R=65$ pixels, $C=-2$). (a): Auto local threshold for a real image with different threshold heights; yellow edges represent the contours of probed regions. (b): The intensity profile along the white lines on the left image; the color filled areas under the curve represent the identified QDs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

$pixel = (pixel > mean - c) ? object : background.$

The Bernsen routine [17,18] uses a user-provided contrast threshold. If the local contrast (max-min) is above or equal to the contrast threshold, the threshold is set at the local midgrey value (the mean of the minimum and maximum grey values in the local window). If the local contrast is below the contrast threshold, the neighborhood is considered to consist of only one class and the pixel is set to object or background depending on the value of the midgrey. The parameter 1 is the contrast threshold. The default value is 15. Any number different from 0 will change the default value.

```
if (local_contrast < contrast_threshold)
    pixel = (mid_gray >= 128) ? object : background
else
    pixel = (pixel >= mid_gray) ? object : background
```

The Contrast routine [19] is based on a simple contrast toggle. It sets the pixel value to either white (255) or black (0) depending on whether its current value is closest to the local maximum or minimum, respectively.

The MidGrey routine [13–16] selects the threshold as the mid-grey of the local greyscale distribution (i.e. $(max + min)/2$). A variation of this method uses the mid-grey - C , where C is a constant. The parameter 1 is the C value. The default value is 0. Any other number will change the default value.

$pixel = (pixel > ((max + min) / 2) - c) ? object : background$

The Niblack routine [20] implements Niblack's thresholding method that uses the following equation for segmentation where the parameter 1 is the k value, the default value is 0.2 for bright objects and -0.2 for dark objects. Any number different from 0 will change the default value. The parameter 2 is the C value. This is an offset with a default value of 0. Any number except for 0 will change its value.

$pixel = (pixel > mean + k * standard_deviation - c) ? object : background$

The Otsu routine [21] implements a local version of Otsu's global threshold clustering. The algorithm searches for the threshold that minimizes the intra-class variance, defined as a weighted

sum of variances of the two classes. The local set is a circular ROI (Region Of Interest) and the central pixel is tested against the Otsu threshold found for that region.

The Sauvola routine [22] Implements Sauvola's thresholding method, which is a variation of Niblack's routine, where the parameter 1 is the k value and the parameter 2 is the r value.

$pixel = (pixel > mean * (1 + k * (standard_deviation / r - 1))) ? object : background$

The Phansalkar routine [23] is a modification of Sauvola's thresholding method appropriate to deal with low contrast images, where mean and stdev are the local mean and standard deviation, respectively. Phansalkar recommends $k=0.25$, $r=0.5$, $p=2$ and $q=10$. k and r are the parameters 1 and 2 respectively, but the values of p and q are fixed.

$pixel = (pixel > mean * (1 + p * \exp(-q * mean) + k * ((stdev / r) - 1))) ? object : background$

Fig. 8 shows the result of the Median method with $R=65$ pixels and $C=2$ for the same image depicted in Fig. 7. The method adapts to the base height variations across the image and provides an accurate evaluation of the QD heights. However, the local sensitivity of the method leads to the detection of intensity variations in the substrate, where no QDs are present. This is shown in more detail in Fig. 9, where the green circles highlight these noisy regions. These defects can be minimized by adjusting the radius and the sensitivity parameters or by size and/or shape filtering.

Another problem that arises is that two or more QDs may overlap and be regarded as a single one, as highlighted by the blue circles in Fig. 9.

Concerning the overlap of large QDs, a separation of these particles is obtained with the *watershed* tool (Fig. 10, blue circles), an implementation of the *watershed* immersion algorithm written by Vincent and Soille [24]. Finally, small particles are eliminated by applying an area size filter (Fig. 10, green circles).

5. Results and discussion

The QD height distribution obtained by the proposed method was fitted with a log-normal distribution, which is one of the distributions most often observed in nature and that describes a large number of physical, biological and even sociological phenomena

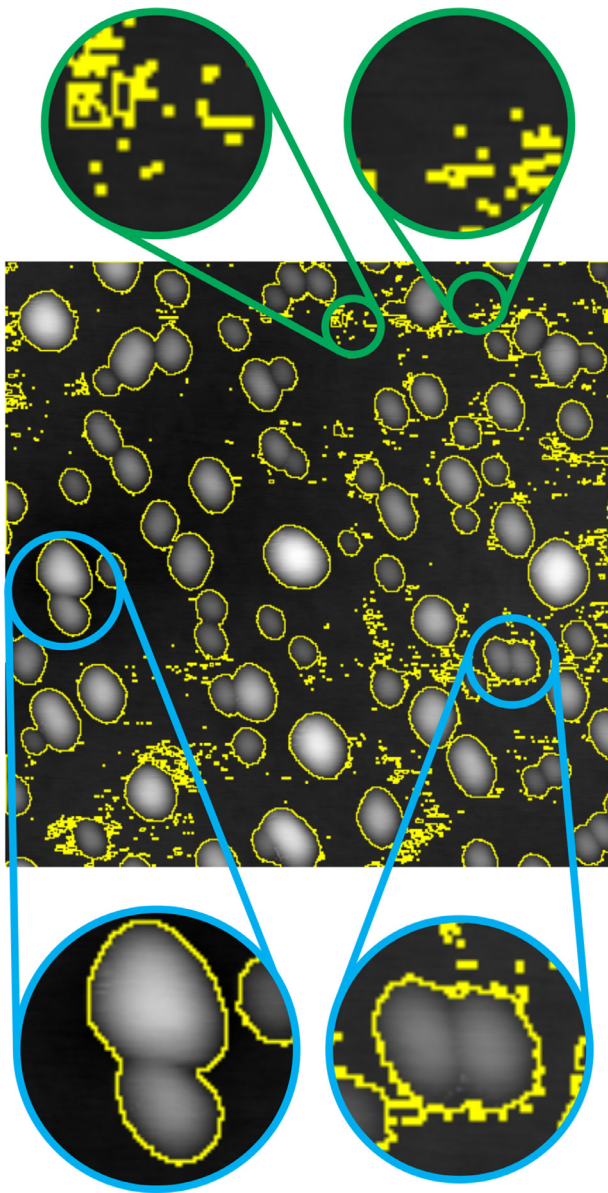


Fig. 9. Image of a surface with QDs obtained by AFM. The green circles highlight small spurious particles in the background. Blue circles highlight touching or coalesced QDs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

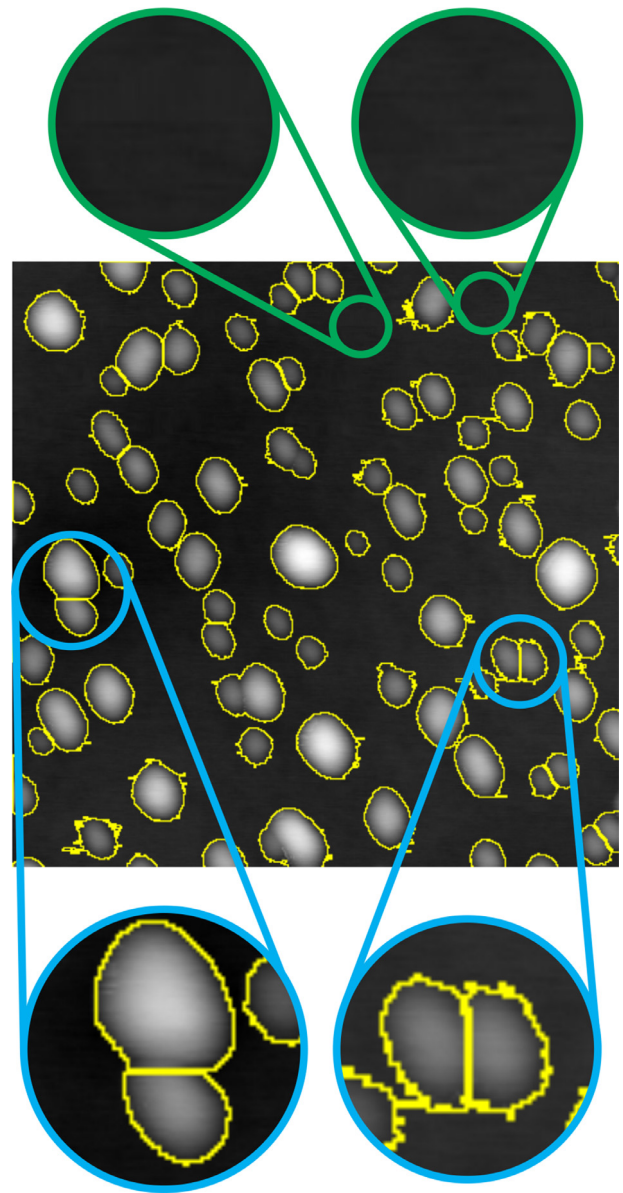


Fig. 10. Image of a surface with QDs obtained by AFM: in blue and green circle, respectively, the separation of particles with the watershed method and the elimination of small particles by applying an area size filter are highlighted. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

[25]. This distribution is commonly observed as a result of various crystallization processes [26,27].

Comparing the results obtained for the same image using the conventional (with a global threshold) and optimized methods shown in Fig. 11(a) and (b), respectively, it is clear that the latter fits much better a log-normal distribution (correlation coefficient R^2 increases from 0.964 to 0.986). The optimized method also detects many more QDs (473 against 410) and leads to a much larger mean QD height (6.3 against 3.9 nm).

In order to verify the accuracy of the developed method, a visual count was performed on the same image and 491 QDs were identified, a value closer to the value detected by the improved method. Fig. 8(b) shows that the local threshold estimated by the proposed method is more accurate for each QD, as the base height adapts to each location.

The median height of the QDs for the conventional and improved methods were used to calculate the theoretical electronic

energy levels using a split operator method in 3D [28], where the strain is taken into account through the InAs bandgap and the QDs are considered disk-shaped. Since the QDs' capping layer is 10 nm thick, the QDs are fully covered, minimizing changes in height in comparison to the free standing ones, probed by AFM. The energy levels were compared with the measured photoluminescence (PL) peak energies.

The PL was performed by focusing an argon ion laser with wavelength of 514.5 nm on a sample kept at 15 K in a cryostat. The PL signal was detected by an InGaAs photodetector after it was dispersed by a monochromator. Synchronous detection was applied.

Fig. 12 shows the observed spectrum, where three peaks can be identified. Peak 1, around 1 eV, is attributed to the emission from the InGaAlAs barrier material, as confirmed by measurements performed on a sample without QDs. The broader emission peaked at 0.73 eV, absent on the sample without QDs, can be fitted by two

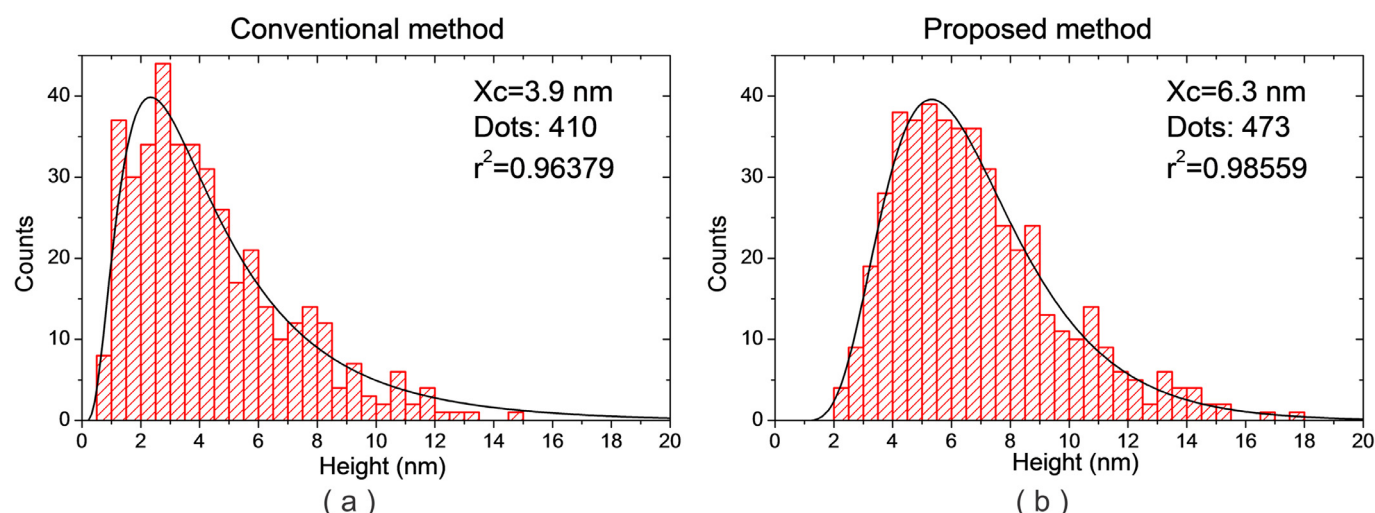


Fig. 11. The QD height distribution determined by the conventional (a) and the proposed (b) methods and fitted with a log-normal distribution.

Table 1
Comparison between transition energies calculated from QD heights and obtained from PL.

Method	Median Height (nm)	Calculated Energies (meV)		PL Experimental Energies (meV)		Error (%)	
		Fund. State	First Excited State	Fund. State	First Excited State	Fund. State	First Excited State
Conventional	3.9	765.8	817.9	699.5	744.6	9.48	9.84
Proposed	6.3	709.9	764.2			1.49	2.63

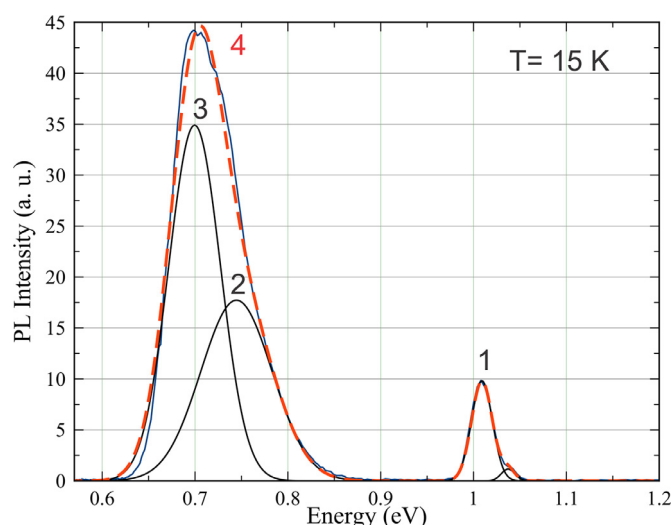


Fig. 12. The blue solid line is the PL spectrum of the investigated sample. The black Gaussians represent the different PL emissions of the sample. Gaussian 1 - peak of the quaternary material = 1009.1 meV. Gaussian 2 - peak of the first QD excited state = 744.6 meV. Gaussian 3 - peak of the QD fundamental state = 699.5 meV. The red dashed line 4 represents the sum of the Gaussians. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

PL emissions (peaks 2 and 3 in Fig. 12) corresponding to transitions departing from the fundamental and first excited electronic states of the QDs. Table 1 summarizes the results.

The calculated transition energies using the QD height determined by both the conventional and proposed threshold methods can be compared with the measured PL peak energies. The relative error in the peak positions decreased from 9.48% to 1.49% for the fundamental state and from 9.84% to 2.63% for the first excited state, when the local threshold is applied, once again indicating that it leads to more accurate results.

6. Conclusion

An improved image processing method for detecting QDs and evaluating their height and density is proposed and tested.

The use of a local threshold, as opposed to the traditional global threshold approach, has proven to be successful in the segmentation of QDs, leading to a more accurate determination of their base heights, and thus of their full height. Touching/overlapping QDs were separated with the well-known *watershed* method. Residual noise due to the local detection characteristics of the local threshold was treated with a size filter.

The proposed method was implemented as a macro in the free Fiji/ImageJ software, and can be easily adapted for other similar applications. The excellent agreement between the obtained results with different experimental data demonstrates that the proposed method is more accurate than other reported ones [29–31] or used in software dedicated to AFM image processing.

It should be pointed out that the achieved improvement in the accuracy in the interpretation of AFM images of QDs has a general character, since the results were reproduced for a large collection of samples.

Acknowledgments

This work was partially supported by CAPES, CNPq, FAPERJ and FINEP. The research is part of the program of the *Instituto Nacional de Ciência e Tecnologia em Nanodispositivos Semicondutores* -DISSE-PUC-Rio.

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